

JOLAN THELAN GOVENDER

BScHons Actuarial Science Candidate | University of the Witwatersrand

jolangovender07@gmail.com | +27 79 490 9096 | [linkedin.com/in/jolangovender](https://www.linkedin.com/in/jolangovender) | jolangt.github.io

PROFESSIONAL SUMMARY

Final-year BScHons Actuarial Science student at Wits with 8 ASSA exemptions and a research focus on machine learning applied to mortality forecasting. Drawn to problems where data and real-world consequences meet, which is what pulled me toward actuarial science and keeps me reading about financial markets and economic history in my own time. Outside of coursework, I tutor over 120 second-year students, which has made me as comfortable explaining a concept clearly as I am deriving it and strengthened my communication skills.

I appreciate all disciplines of Actuarial Science but I am particularly interested in applying my skills and knowledge in the long-term and short-term insurance spaces. Helping people achieve their financial goals, life aspirations and protecting businesses that people worked hard to build is what drove me to choose these specialisations within the actuarial disciplines. I am dedicated towards achieving my FASSA designation and completing all my actuarial exams.

EDUCATION

University of the Witwatersrand

BScHons – Actuarial Science (2026)

2026 – Present

- Actuarial Risk Management (A311)
- Life Insurance (F102)
- Health, Social and Employee Benefits (F108)

BSc – Actuarial Science & Mathematical Statistics

2023 – 2025

Star College Boys High School, Durban

National Senior Certificate – 7 Distinctions, 88% Average

2018 – 2022

ASSA EXEMPTIONS

- | | |
|---|---|
| • A112 (Business Economics) | • A213 (Contingencies) |
| • A113 (Business Finance) | • A214 (Actuarial Economic Modelling) |
| • A211 (Financial Mathematics) | • N111 (Foundation Actuarial Professional Practice) |
| • A212 (Risk Modelling & Survival Analysis) | • N211 (Communications) |

RESEARCH

Mortality Forecasting Using Recurrent Neural Networks

BScHons Research Project · University of the Witwatersrand

2026 – Present

Constructed and evaluated Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) recurrent neural network architectures for long-horizon mortality forecasting, benchmarked against the classical Lee-Carter model on a 10-year test set.

Key findings: Lee-Carter substantially outperformed both RNN architectures overall, with RNNs failing to capture mortality improvement trends. A notable exception was Japanese female mortality, where LSTM outperformed Lee-Carter.

Findings speak directly to concerns raised in actuarial literature around the reliability and stability of neural network approaches to mortality projection.

EXPERIENCE

Tutor – Actuarial Science (2nd Year)

University of the Witwatersrand

2026 – Present

- Support 120+ second-year students across Financial Mathematics, Business Applications, and Life Contingencies through weekly tutorials.

MiWay Actuarial Hackathon

MiWay Insurance

2026

- Modelled the risk cost and office premium for 500 000 policies, simulating multiple years in a competitive insurance market, and placed 3rd out of the competing teams.
- Created Light Gradient Boosting-Machine models, split into frequency and severity, to accurately capture the interactions between different rating factors.

TECHNICAL SKILLS

- | | | |
|------------------------------|---------------------------|--------------------|
| • R (Advanced) | • Financial Mathematics | • Machine Learning |
| • Microsoft Excel (Advanced) | • Risk Theory & Reserving | • Neural Networks |
| • Financial Modelling | • Life Contingencies | |
| | • Statistical analysis | |

LANGUAGES

- English (Native) | Afrikaans (Working Proficiency)

AWARDS & INTERESTS

- Good Fellowship Award – Star College Boys High, 2022
- Bronze Medal – Golden Climate Environmental Project Olympiad
- 3rd Place – MiWay Actuarial Hackathon 2026
- Interests: Golf, Padel, Weight-lifting, Financial Markets, Chess

REFERENCES

Rivaj Maharaj (FASSA)

Lecturer, University of the Witwatersrand

rivaj.maharaj@wits.ac.za | +27 84 073 3114

Rendani Mbuva

Associate Professor, Actuarial Science, University of the Witwatersrand

rendani.mbuva@wits.ac.za | +27 72 087 2571